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1.

Step 1: We use the Laplacian kernel of the form:

−1	−1	−1
−1	8	−1
−1	−1	−1

And slide the kernel through the image to get the Laplacian image.

Step 2: We then add the Laplacian image to the original image to get the Laplacian Sharpened image.

Step 3: We then use the Sobel kernels:

−1	−2	−1
0	0	0
1	2	1

−1	0	1
−2	0	2
−1	0	1

To calculate the x-directions & y-directions gradient, we then use the Sobel kernels to get the Sobel Gradient.

Step 4: We then use a 5x5 median filter to smooth out the Sobel Image, giving us a "Sobel Smooth Image".

Step 5: We then multiply the Laplacian Image with the Sobel Smooth Image, giving us the masked image.

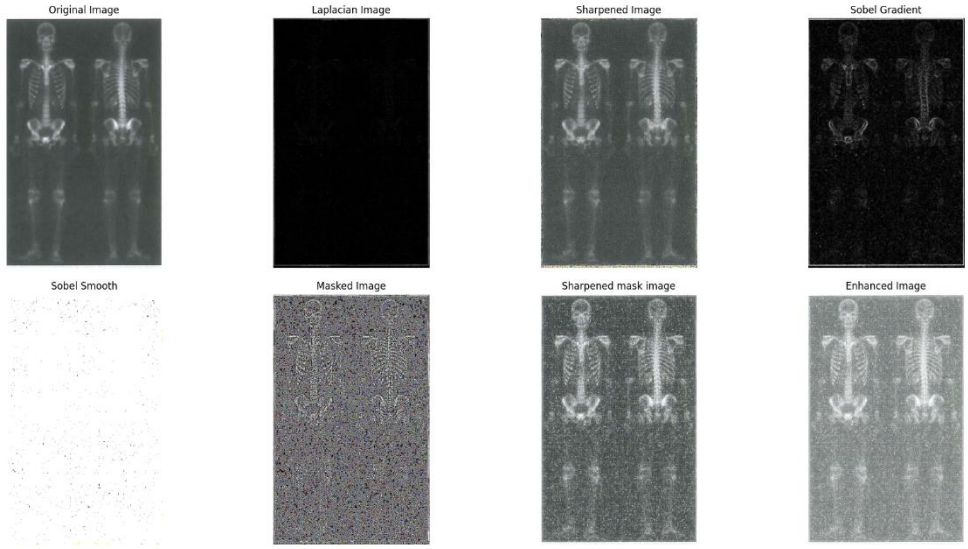
Step 6: We then add the masked image to the original image, giving us the sharpened image.

Step 7: We then take the sharpened image and use the power-law transformation:

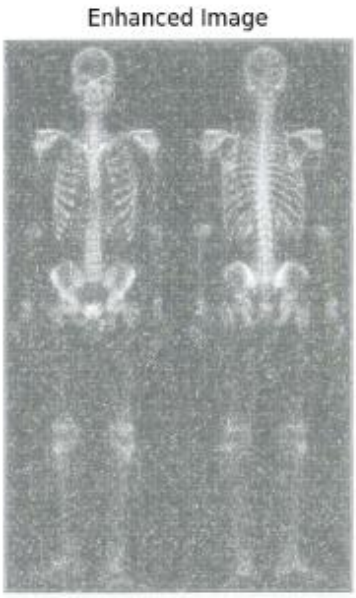
$$y = cr^{\gamma}$$

with parameters c = 1 & γ = 0.5 to enhance the brightness of the overall image to get the Final Enhanced Image.

All the Procedures look like this:



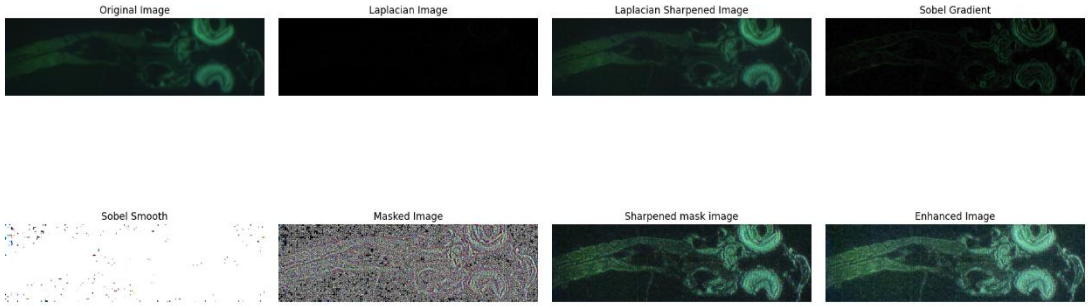
And the result is:



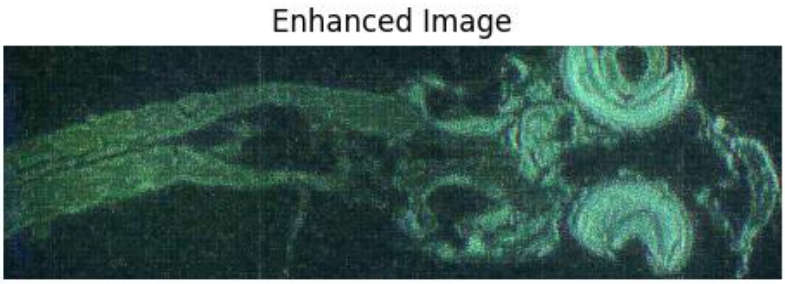
2.

The steps used to enhance “fish.jpg” are the same as those of the “bodybone.bmp”.

All the Procedures look like this:



And the Final Result is:



3. During the processing of the image ‘bodybone.bmp’.We can view that applying only the Laplacian filter produced rather poor results as the image became nearly invisible. However, when the Laplacian-filtered image was added back to the original to form a sharpened image, the edges and fine details became noticeably more distinct. The sharpened mask image, generated through Sobel and Laplacian filtering revealed clearer edges than the original image but also introduced more noise. Lastly, applying a power-law transformation increased the brightness of the image.

   A similar outcome was observed in the processing of ‘fish.jpg’. The sharpened image showed a much better result compared to the one processed with only the Laplacian filter. The Sobel Gradient produced sharper edges than the sharpened image, while the Sobel Smooth version barely revealed any visible features. The sharpened mask image, compared to the original, displayed much clearer edges and finer details, and applying the power-law transformation further enhanced the image brightness.